Inference of proportions.

1) Single proportion

$X \longleftrightarrow p$.

2) Two proportions (diff of proportions)

X_1 X_2

p_1 p_2

Theory on single proportion

Model:

$X \sim \text{binom}(n, p)$

n = no. of trial. → known

p - prob of success → unknown

Check BINS assumption.

p = population proportion. → fixed

$= \frac{\text{no. of success}}{\text{total no. of outcomes}}$

$= \frac{\text{no. of success}}{N}$

parameter of interest.

P.E : sample proportion

$\hat{p}$ → not fixed

$$\hat{p} = \frac{\text{no. of success in sample}}{\text{sample size.}}$$

↳ point estimate for p.

↳ random variable.

Sampling dist of p.e ?

$E(\text{p.e}) = \text{parameter}$; s.e $(\hat{p})$

$E(\hat{p})$

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

sampling dist of $\hat{p}$.

$\hat{p} = \frac{1}{n}$ no. of successes.

$= \frac{1}{n} \Sigma I_s$.

$\underbrace{\quad\quad}$ average of indicator.

Confidence Interval

general formula: p.e $\pm$ c.v x s.e

100c% CI of p:

$$\hat{p} \pm \text{c.v} \times \sqrt{\frac{p(1-p)}{n}}$$

Since S.e depends on p which is unknocon.

We will have to find an approximation or estimate S.e.

$$S.e = \sqrt{\frac{p(1-p)}{n}}$$

$$\hat{S.e} = S.e_{adj} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

C.I for p:

$$\hat{p} \pm c.v \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$C.V = qnorm\left(C + \frac{(1-C)}{2}\right)$ → depends on C or α.

$1-\frac{\alpha}{2}$

$qdist\left(C + \frac{(1-C)}{2}\right)$

↓

sampling dist

$C.V = qnorm\left(C + \frac{(1-C)}{2}\right)$

↳ Z

$\hat{p} \pm qnorm\left(C + \frac{(1-C)}{2}\right) \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Inference on single proportions

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1) Model check : $X \sim$ Binomial
↳ BMVS.

parameter : p.

$X \sim \text{Binom}(n, p)$.

n = no. of trials (known)

p = prob of success / pop prop
(unknown).

P.E : sample prop = $\hat{p}$

Sampling dist of $\hat{p}$:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

sd of binom rv $= \sqrt{np(1-p)}$

$E(\hat{p}) = p$

$s.e(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$ ← unknown

$s.e_{adj}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ (Wald adj)

AC adj : Agresti-Coull adj

CI for p:

General formula: p.e ± c.v × s.e

$$\hat{p} \pm c.v \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$c.v = qnorm\left(1-\frac{\alpha}{2}\right)$ ← removing the upper tail

$\left(c+\frac{1-c}{2}\right)$

↳ z-score of the endpoints.

100 c.l. CI for p:

$$\hat{p} \pm qnorm\left(1-\frac{\alpha}{2}\right) \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Hypothesis testing:

1) Model statement & stating the hypotheses:

Model: $X \sim \text{Binom}(n, p)$

(BINS assumption)

Hypotheses: $H_0: p \geq \text{value}$

$H_0: p = \text{value}$

$H_0: p \leq \text{value}$

$H_A: p \neq \text{value}$.

Example: Muriel

$n = 8$

$X =$ num of cups Muriel gets right. $\sim \text{Binom}(8, p)$

Binomial model for hypothesis p.

$H_0: p = 0.5$ (guessing at random)

$H_a: p > 0.5$ (guessing with an ability).

$H_0: X = 4$

$H_a: X > 4$

Under H_0: $X_{H_0} \sim Binom(8, 0.5)$

$x = 6$

p-val = Prob of getting an obs value as or more extreme.

$P(X \geq 6) = pbinom(5, 8, 0.5, lower.tail = F)$

$= 0.144$

$\alpha = 0.05$

p-val $> \alpha \Rightarrow$ We fail to reject H_0.

$X = 8$

p-val = pbinom(7, 8, 0.5, lower.tail = F)

$= 0.004 < \alpha$

We reject H_0.

Inference on single prop

$X \sim \text{Binom}(n, p)$

100c% C.I for p:

p.e ± c.v × s.e

c.v ~ sampling dist of p.e

$$\hat{p} \pm \text{qnorm}\left(c + \frac{1-c}{2}\right) \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

qnorm ~ sampling dist of $\hat{p}$ is normal

$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ ~ wald adj of s.e

Hypothesis testing for p:

1) $H_0: p = \text{value}$ ⟸

$H_A: p \begin{matrix} < \\ > \\ \neq \end{matrix} \text{value}$

Binomial model for testing:

$X \sim Binom(n, p)$

$X_{H_0} \sim Binom(n, value)$ ←

↳ p.e : no. of success = x

pval = pbinom(x_{obs}, n, p)

or 1 - pbinom(x_{obs}, n, p)

or lower tail prob + upper tail prob.

Example: Psychic example in the lecture notes

Normal model of testing:

p.e: $\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$

unknown

test statistic for p.e:

$$z = ts = \frac{\hat{p} - p_{H_0}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$$

null dist.

p-val: pnorm(−z)

or 1 − pnorm(z)

or 2 × pnorm(z, lower.tail = F)

p-val $< \alpha$ $\rightarrow$ Reject H_0

p-val $\geq \alpha$ $\rightarrow$ we fail to reject H_0.

Inference on diff on prop.

$X_1 \sim \text{Binom}(n_1, p_1)$ $X_2 \sim \text{Binom}(n_2, p_2)$

p_1 $\rightarrow$ parameter $\leftarrow$ p_2

parameter : $p_1 - p_2$

individual p.e : $\hat{p}_1$ $\hat{p}_2$

p.e : $\hat{p}_1 - \hat{p}_2$ (parameter $= p_1 - p_2$)

$E(\text{p.e}) = \text{parameter}$.

$$E(\hat{p}_1 - \hat{p}_2) = E(\hat{p}_1) - E(\hat{p}_2)$$

$$= p_1 - p_2$$

$$\hat{p}_1 \sim N\left(p_1, \sqrt{\frac{p_1(1-p_1)}{n_1}}\right)$$

$$\hat{p}_2 \sim N\left(p_2, \sqrt{\frac{p_2(1-p_2)}{n_2}}\right)$$

$$\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

p.e

sampling dist

$$E(\hat{p}_1 - \hat{p}_2) = p_1 - p_2$$

$$s.e(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

$$s.e\,adj(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

wald adj.

C.I for $p_1 - p_2$:

$$p.e \pm c.v \times s.e$$

$$(\hat{p}_1 - \hat{p}_2) \pm qnorm\left(1 - \frac{\alpha}{2}\right) \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Hypothesis testing for $p_1 - p_2$

Parameter: $p_1 - p_2$

p.e: $\hat{p}_1 - \hat{p}_2$

sampling dist: $N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$

1) Model statement & stating the hypothesis.

X_1 & X_2 → binomial.

$H_0: p_1 - p_2 = 0$

$H_a: p_1 - p_1 \begin{matrix} > \\ < \\ \neq \end{matrix} 0$

Since hypo test always assumes that H_0 is true.

$$H_0: p_1 - p_2 = 0 \Rightarrow p_1 = p_2 = p$$

Sampling dist under H_0:

$$N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

under H_0 $\Rightarrow$ $$N\left(0, \sqrt{\frac{p(1-p)}{n_1} + \frac{p(1-p)}{n_2}}\right)$$

$$s.e(\hat{p}_1 - \hat{p}_2) = \sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

p unknown (since, p is unknown)

$X_1 \sim \text{Binom}(n_1, p)$

$X_2 \sim \text{Binom}(n_2, p)$

$\hat{p}_1$ & $\hat{p}_2$ are always distinct

p = global proportion of pop.

↳ $\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$ (global prop of samples)

$$s.e_{adj} = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

2) test statistic & null dist

$$ts = \frac{p.e - \text{parameter}_{H_0}}{s.e_{adj}}$$

$$ts = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)_{H_0}}{S.E_{adj}(\hat{p}_1 - \hat{p}_2)}$$

$(p_1 - p_2)_{H_0} = 0$

$S.E_{adj}(\hat{p}_1 - \hat{p}_2)$

$$z(ts) = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$\sim N(0,1)$

null dist

test statistic

3) p-val = pnorm(-z) (left-tailed)

or pnorm(z, lower.tail=F)

(right-tailed)

or 2 × pnorm(|z|, lower.tail=F)

∵

normal is symmetric.

generalized p-val: pnorm(|z|, lower.tail=F)

4) conclusion:

p-val with α

p-val $< \alpha$ ⇒ Reject H_0

p-val $\geq \alpha$ ⇒ we fail to reject H_0.